This covers zero-shot use of text features as linear classifiers, standard linear probing, and the affine last layer in most neural classifiers. If h is non-linear in a neural network, we absorb the layers preceding the linear layer g into the encoder ($\tilde{f} = g \circ f$) and analyze the resulting affine layer. The definition below keeps the readout h general to allow future extensions beyond linear heads, but all results in this paper consider the linear case, without such restrictions, a high-capacity readout could make any injective encoder appear compositional by memorization.

Definition 3 ((Linearly) compositional model)**.** An encoder $f : \mathcal{X} \to \mathcal{Z}$ is *compositional w.r.t.* $\mathcal{C}$ if there exists $h : \mathcal{Z} \to \mathbb{R}^{k \times n}$ such that, for all $\boldsymbol{c} \in \mathcal{C}$ and all $i \in [k]$,

$$c_i = \operatorname*{argmax}_{j \in [n]} h(f(\boldsymbol{x_c}))_{i,j} \,. \tag{2}$$

It is *linearly compositional* if h can be taken affine $h(\boldsymbol{z}) = \boldsymbol{W}\boldsymbol{z} + \boldsymbol{b}$. We refer to h as the *readout*.

3.3. Compositional generalization and desiderata

Given the ingredients (concept space $\mathcal{C}$, encoder f, and training-support family $\mathcal{T}$), we now define a learning rule A and state three desiderata for compositional generalization: *divisibility*, *transferability*, and *stability*. We emphasize that these desiderata are on the NN-based models that exhibit generalization, as defined below, not on the representations, as studied in disentangled representation learning.

Considering training. We view a learning algorithm as a map

$$A : \; D_T \;\mapsto\; h_T, \qquad h_T \in \mathcal{H} \subseteq \{h : \mathcal{Z} \to \mathbb{R}^{k \times n}\},$$

from a dataset supported on $T \subseteq \mathcal{C}$ to a readout in a chosen hypothesis class. In practice, A is typically (stochastic) gradient descent on a cross-entropy or contrastive objective, covering contrastive vision-language encoders (e.g., CLIP, SigLIP), standard supervised classifiers, and linear probes on self-supervised vision encoders like DINO.

Desiderata for compositional generalization. Suppose we train a downstream readout $h_T = A(D_T)$ on some $T \in \mathcal{T}$. What should h_T satisfy? We argue for three practically-motivated properties.

First, every combination of concept values should be *classifiable* by the readout: for any $\boldsymbol{c} \in \mathcal{C}$, the corresponding region of the representation space of f is nonempty: there exists at least one $\boldsymbol{z}$ that h_T assigns the concept values $\boldsymbol{c}$. Otherwise, generalization to the full grid is impossible. We refer to this property as *Divisibility*.

Desideratum 1 (Divisibility)**.** A compositional model (Definition 3) can only exist if the readout has sufficient capacity to represent every concept combination. That is, for a readout $h : \mathcal{Z} \to \mathbb{R}^{k \times n}$, every concept tuple must have a nonempty decision region:

$$\begin{aligned} &\forall\, \boldsymbol{c} \in \mathcal{C} : \bigcap_{i=1}^{k} \mathcal{R}_{i,c_i}(h) \neq \varnothing, \\ &\text{where } \; \mathcal{R}_{ij}(h) = \{\boldsymbol{z} \in \mathcal{Z} : \arg\max_{j' \in [n]} h(\boldsymbol{z})_{i,j'} = j\}. \end{aligned} \tag{3}$$

Divisibility is necessary but not sufficient: it guarantees that the space is divisible, but does not imply that the readout will be correct. We therefore ask that, for every training set, the learned readout transfers to the full grid; we refer to this as *Transferability*.

Desideratum 2 (Transferability)**.** For every $T \in \mathcal{T}$, the trained readout $h_T = A(D_T)$ correctly classifies all possible combinations of the concept space:

$$\forall\, \boldsymbol{c} \in \mathcal{C},\; \forall\, i \in [k] : \quad \operatorname*{argmax}_{j \in [n]} h_T\big(f(\boldsymbol{x_c})\big)_{i,j} = c_i. \tag{4}$$

Note that Transferability implies Divisibility. We state Divisibility explicitly because it highlights a capacity requirement: the embedding space must be able to represent all concept combinations.

Third, consider readouts learned from different valid supports $T \in \mathcal{T}$. Divisibility and Transferability do not say anything about the behavior of the classification decisions. Intuitively: if an input depicts a "cat", retraining on another valid support should not flip the preference to "dog" or push the prediction toward near-indifference. We refer to this as *Stability*.

Desideratum 3 (Stability)**.** For any $T, T' \in \mathcal{T}$, any point $\boldsymbol{x} \in \mathcal{X}$, and any $i \in [k]$, the per-concept posteriors agree across supports:

$$\begin{aligned} p_i^{(T)}(j \mid f(\boldsymbol{x})) &= \frac{\exp(h_T(f(\boldsymbol{x}))_{i,j})}{\sum_{k=1}^{n} \exp(h_T(f(\boldsymbol{x}))_{i,k})}, \\ p_i^{(T)}(\cdot \mid f(\boldsymbol{x})) &= p_i^{(T')}(\cdot \mid f(\boldsymbol{x})). \end{aligned} \tag{5}$$

We view Stability as an idealization: once the training support is sufficiently informative, retraining on any other valid support should not change the model's per-concept predictions (and ideally its calibrated posteriors). In practice, this may only hold approximately; relaxing Stability to allow small distributional deviations across supports is a natural direction for future work. We discuss the role of Stability and what can fail without it in Section G.

Defining compositional generalization. We now tie the ingredients into a single tuple $\Pi = (f, \mathcal{H}, A, \mathcal{T})$, which we use as the object that specifies the entire compositional-generalization setup: the encoder, the readout class, the learning rule, and the family of valid training supports. We specify compositional generalization as a process of learning readouts that generalize over *all* $T \in \mathcal{T}$ and satisfy Desiderata 1–3.

Definition 4 (Compositional generalization)**.** $\Pi = (f, \mathcal{H}, A, \mathcal{T})$ *exhibits compositional generalization* if, for

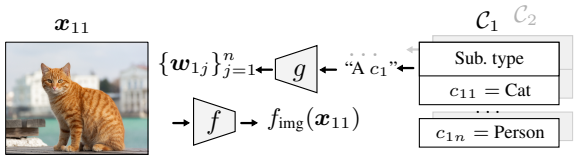

Figure 4. **Instantiating the framework with CLIP-like embedding models for analysis.**

every $T \in \mathcal{T}$ with $h_T = A(D_T)$, Divisibility (Desideratum 1) and Transferability (Desideratum 2) hold on the full grid, and the posteriors are Stable across valid retrainings (Desideratum 3) for all pairs $T, T' \in \mathcal{T}$. We say that Π exhibits *linear compositional generalization* when the readout hypothesis class is linear.

We illustrate the relationship between (linear) models and their compositional counterparts in Fig. 3. In practice one could consider relaxed or average-case variants; however, we here are interested in "ideal" representations that support compositional generalization under any data sample.

3.4. Instantiating the framework with CLIP

We instantiate the framework in the dual-encoder, vision-language setting in the style of CLIP models: images and texts are embedded into a shared space and trained to align, with captions acting as noisy descriptions of concept tuples.

Encoders. Let $f : \mathcal{X} \to \mathcal{Z}$ be the image encoder and $g : \mathcal{Y} \to \mathcal{Z}$ the text encoder. At inference both are typically ℓ_2-normalized so that inner products are cosine similarities: $\|f(\boldsymbol{x})\| = \|g(\boldsymbol{y})\| = 1$.

Prompts as linear probes. Zero-shot classification uses text features as linear classifiers. For each concept $i \in [k]$ and value $j \in [n]$, we can choose a prompt $p_{i,j}$ (e.g., "a photo of a cat") and define a probe vector $\boldsymbol{w}_{i,j} := g(p_{i,j}) \in \mathcal{Z}$. Stacking these gives a readout

$$h(\boldsymbol{z}) = \left[\ \boldsymbol{w}_{i,j}^{\intercal}\boldsymbol{z}\ \right]_{i,j} \in \mathbb{R}^{k\times n}.$$

Here f is the representation model, while h is a linear readout whose weights come from the text encoder. Training in CLIP-like models can be viewed as learning a readout model where the *same* set of text-derived probes serves across many images; prompts often mention only parts of an image, so the system is implicitly asked to recognize objects and attributes regardless of which other concepts co-occur. We illustrate this process in Fig. 4; for a high-level schematic, see Fig. 1 in the introduction.

The question we study. Given a concept space $\mathcal{C}$, what structure must $\boldsymbol{z} = f(\boldsymbol{x_c})$ have so that a single set of probes $\{\boldsymbol{w}_{i,j}\}$ (whether fixed by g or learned as linear probes) satisfies our desiderata (Desiderata 1–3) on the full $\mathcal{C}$? In other words, what constraints does zero-shot, probe-based classification place on the geometry of image representations in order to support compositional generalization?

4. Implications of compositionality on representations

We now ask what our desiderata imply for representations in common training regimes, both as necessary constraints and as sufficient conditions. Three questions guide the section:

Q1 (§4.1) *Geometry under GD with CE/BCE and stable transfer.* If A is gradient descent under binary cross-entropy, and Π exhibits compositional generalization (Def. 4) across a family of supports $\mathcal{T}$, what structure is *necessary* for f (and the linear readout h)? $\to$ We show additive (linear) factorization with orthogonal concept directions under natural $\mathcal{T}$.

Q2 (§4.2) *When are these geometric conditions sufficient?* In the same binary GD+CE regime, do linear factorization and cross-concept orthogonality also *suffice* for compositional generalization? $\to$ Yes, establishing that the geometry is both necessary and sufficient.

Q3 (§4.3) *Minimal dimension of linearly compositional models.* Assuming divisibility holds (Desideratum 1) and a linear (affine) readout h, what is the smallest d so that correct per-concept predictions are possible over all n^k tuples? $\to$ With affine readouts, $d \geq k$ is necessary.

4.1. Geometry of f under common training settings

We instantiate A as gradient descent on the binary cross-entropy (logistic) loss. As in Section 3.4, the readout h is linear in the embedding $\boldsymbol{z_c} = f(\boldsymbol{x_c})$ (using either text-encoder-derived probes or learned linear heads). Under these assumptions, the representation space $\mathcal{Z}$ must exhibit both linearity and cross-concept orthogonality. This conclusion holds under at least two validity regimes: (1) when more than half of all possible combinations are observed, $|T| = 2^{k-1} + 1$, and (2) when only a small, carefully chosen set of datapoints is observed, $|T| = 1 + k$.

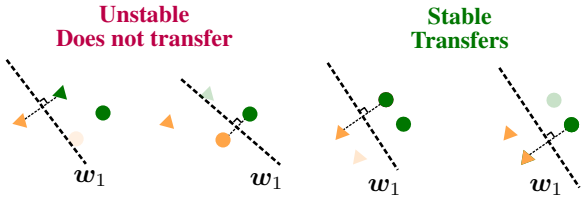

Figure 5. **Stable and unstable examples of feature representations.** The top panel shows an unstable configuration, where depending on the sample, the readout either does not transfer or unstably. Bottom panel shows a stable configuration.

Proposition 1 (Compositional generalization implies linear factorization)**.** Let $\Pi = (f, \mathcal{H}, A, \mathcal{T})$ be the tuple instantiated in Definition 4, with linear heads $\mathcal{H}$ and A given by GD+CE. Suppose that the training sets follow random sampling with validity rule $R(T) = 1$ if $|T| = 2^{k-1} + 1$. Assume Desiderata 1–3 are satisfied. Then, under the binary grid $\mathcal{C}_i = \{0, 1\}$ with $\mathcal{Z} = \{\boldsymbol{z_c} : \boldsymbol{c} \in [2]^k\} \subset \mathbb{R}^d$, there exist $\{\boldsymbol{u}_{i,0}, \boldsymbol{u}_{i,1} \in \mathbb{R}^d\}_{i=1}^k$ such that for every $\boldsymbol{c} \in [2]^k$ the following holds:

1. *(Linearity)* $\boldsymbol{z_c} = \sum_{i=1}^k \boldsymbol{u}_{i,c_i}$.
2. *(Cross-concept orthogonality)* $(\boldsymbol{u}_{i,1} - \boldsymbol{u}_{i,0}) \perp (\boldsymbol{u}_{j,1} - \boldsymbol{u}_{j,0})$ for all $i, j \in [k]$ with $(i \neq j)$.

Proof sketch. GD+CE converges to a max-margin SVM (Soudry et al., 2024). Stability implies consistent weight differences; max-margin with varying training sets makes each point a support vector, yielding prediction invariance when other concepts change. Max-margin geometry then implies flipping a concept produces an additive shift. Because of the SVM solution, weights for each concept are proportional to the segment connecting positive and negative classes, from which follows orthogonality. Minimal dataset setting can be argued similarly by carefully constructing datasets with size $1 + c$ to contain only pairs of datapoints differing only in one concept value.

Intuitively, linear factorization means that a combination space of n^k elements can be explained using only $n \cdot k$ factors. The orthogonality condition says that factors of concept values belonging to different concepts (e.g., "red" and "square") are orthogonal to each other, but no requirement is placed on the factors of concept values belonging to the same concept (e.g., "red" and "blue"). We illustrate the stable and unstable examples of feature representations in Fig. 5. Additionally, we note that linear factorization in itself is not trivial: the fact that n^k datapoints can be explained using $n \cdot k$ factors does not have to hold for any linearly compositional model. We illustrate this with examples in Section H.4.

The datapoint requirement can be interpreted as operating in either (i) a minimal-learning regime for extrapolating to the whole grid (as in Compositional Risk Minimization framework (Mahajan et al., 2025)), where $|T| = 1 + k(n-1)$ suffices to extrapolate to the whole grid, or (ii) a large-sample regime in which random sampling yields near-complete coverage of the concept space.

Empirical evaluation on synthetic data. The necessary conditions discussed so far concern models that support compositional generalization. Neural networks trained from scratch, even on large-scale datasets, may not exhibit this structure. Prior work provides evidence that this structure can emerge under standard classification losses, without explicit pressure to generalize compositionally (Uselis et al., 2025), but those results are limited to neural networks in a two-concept setting. We therefore conduct synthetic experiments to evaluate to what extent linearity (measured by (7)) and orthogonality emerge under standard classification losses, and find positive evidence, especially as the number of concepts increases (see Figs. 38 and 39, and setup in Section I.6). Finally, in Section F.2, we analyze an idealized, but practically desired, setting where a model's logits are exactly equal to α for the matches, and β otherwise, i.e. $(h(\boldsymbol{z}) \in \{\alpha, \beta\})$, and show that this implies strong dimensionality requirements, as well as additive representations (up to projection) that exactly preserve all probe scores (Proposition 7).

Takeaway §4.1. Under gradient descent with cross-entropy loss, compositional generalization with stable transfer requires linear factorization and orthogonality of cross-concept factors.

4.2. Sufficiency of linear and cross-concept orthogonal f

In the same regime as Proposition 1, the converse holds: if embeddings are linearly factorized with cross-concept orthogonality, then GD+CE trained on sufficiently informative supports yields transferability and stability on the full concept space (Proposition 4 and section D.1). This closes the loop for that regime: the geometry is both necessary and sufficient.

Beyond the binary orthogonal case, sufficiency can be achieved, though not necessarily with SGD: if factors are recoverable, one can reconstruct the representations (Section D.2) and construct classifiers that generalize, provided the original representation space admits a linearly compositional model (i.e., satisfies Divisibility). This extends Uselis et al. (2025) to the multi-concept setting. We summarize the sufficiency results in Proposition 2.

Proposition 2 (Sufficiency summary (informal))**.** Under linear readouts, two sufficiency statements hold.

1. *Binary necessity–sufficiency case.* Under the binary grid with GD+CE, if embeddings decompose as $\boldsymbol{z_c} = \sum_i \boldsymbol{u}_{i,c_i}$ with cross-concept orthogonality, then any training set with $|T| = 2^{k-1} + 1$ (or a cross dataset of size $1 + k$) suffices: the learned readout recovers every concept value on the full grid (transferability) and is invariant across valid T (stability). Combined with Proposition 1, the geometry is both necessary and sufficient (Proposition 4).
2. *General constructive case.* In the multi-valued case, recoverable linear factors are sufficient to construct concept readouts in principle (Section D.2).

We believe that only the first case is of practical interest, since it assumes standard GD+CE training; together with the necessary conditions, it reinforces that linear, cross-concept